$$x = \sqrt{m}\, y.$$

In terms of *x*, the Lagrangian becomes

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \tag{10.3}$$

The constant ω is defined as $\omega = \sqrt{\frac{k}{m}}$ and happens to be the frequency of the oscillator.

By making this change of variables, we can describe every oscillator in exactly the same form. In this form, oscillators are distinguished from each other only by their frequency ω.

Now, let's use Lagrange's equations to work out the equations of motion. For this one-dimensional system, there is only one Lagrange equation, namely

$$\frac{\partial L}{\partial x} = \frac{d}{dt}\frac{\partial L}{\partial \dot{x}}. \tag{10.4}$$

Carrying out these operations on Eq. 10.3, we find that

$$\frac{\partial L}{\partial \dot{x}} = \dot{x}. \tag{10.5}$$

This is called the canonical momentum conjugate to *x*. Differentiating with respect to time gives

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \ddot{x}, \tag{10.6}$$